

at 100% load speaks for itself. Along with the modular cable design, you will feel safe installing just about anything inside the cabinet.

Outrageous

The Corsair RM1000i works well for this segment too. You can use the Corsair RM1000x as a replacement as it has similar specs and ratings. If however, 1000 watts is not enough for you, the Corsair AX1500i or Cooler Master MASTERWATT MAKER 1500 should more than suffice.

What to look for in the spec-sheet

First things first – the power rating. Make a calculation of how much power your system will draw when under full load. Since most PSUs operate between 80%-90% efficiency (check this on the spec-sheet), multiply the number with 1.2. You should keep a buffer of about 30% for when you might want to add more components in the future and to make sure that the PSU itself doesn't operate under a full load and can handle the startup power draw with ease. So, if all components are going to draw 400 watts of energy together, you should get $((400 \text{ watts} \times 1.2) + (30\% \text{ of } 400 \text{ watts})) = (480 + 120) \text{ watts} = 600 \text{ watts}$ of power supply just to be on the safe side.

We have already told you about the types of cables that a PSU provides for powering various components. Make sure your PSU has enough of them. For example, if your PSU has only 2 SATA power cables but you plan on having more than 2 SATA devices, you would need to find alternate ways (like converting a MOLEX output to a SATA power output pin by using an adapter) and it may not always be possible.

The most important thing to look out for is safety. PSUs are marked with protection ratings. Some that you can find are:

1. OVP – Over Voltage Protection
2. UVP – Under Voltage Protection
3. OPP/OLP – Over Power Protection / Over Load Protection (they are basically the same)
4. OTP – Over Temperature Protection
5. SCP – Short Circuit Protection

All of the above immediately cut the power supply to the PSU itself and/or rest of the components when any parameter crosses defined limits. Look for these in the spec-sheet. Steer clear from products without these protection technologies mentioned clearly. MTBF, which stands for Mean